

ROMAN, Romania

WMO: 151110

Lat: 46.9692N

Lon: 26.9119E

Elev: 713

StdP: 14.32

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.2	(c) 8.8	(d) 1.5	(e) 6.1	(f) 5.4	(g) 6.5	(h) 7.9	(i) 10.1	(j) 30.6	(k) 21.5	(l) 26.3	(m) 19.5	(n) 2.8	(o) 0	(p) 0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	88.4	69.1	85.6	68.1	83.0	67.1	72.3	82.8	70.7	81.0	69.2	79.2	6.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.4	110.9	76.1	67.5	103.9	74.7	65.9	98.1	73.1	36.5	83.4	35.0	80.9	33.7	79.2	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	18.6	16.3	-3.4	93.3	6.6	2.9	-8.1	95.4	-12.0	97.1	-15.7	98.7	-20.6	100.9
			-3.7	75.4	6.6	2.2	-8.5	76.9	-12.4	78.2	-16.1	79.4	-20.9	81.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.8	26.7	31.9	39.6	50.8	60.4	67.8	69.8	69.4	61.1	49.9	39.1	29.4		
	DBStd	17.22	8.33	9.76	8.82	7.65	6.48	5.24	4.67	5.38	7.64	7.52	9.32	9.85		
	HDD50	2732	723	506	330	85	7	0	0	0	4	96	344	637		
	HDD65	6071	1188	925	786	427	166	32	14	22	162	469	778	1102		
	CDD50	2644	0	1	9	108	331	533	614	602	338	92	16	0		
	CDD65	507	0	0	0	1	25	114	162	159	45	1	0	0		
	CDH74	4667	0	0	0	23	292	980	1364	1528	465	15	0	0		
	CDH80	1358	0	0	0	1	44	240	400	551	122	0	0	0		
(14) Wind	WSAvg	6.1	6.4	6.5	7.4	7.5	6.4	5.7	5.0	5.0	5.4	5.5	6.1	6.3		
(15) Precipitation	PrecAvg	19.9	0.7	0.7	0.8	1.9	2.7	3.2	3.1	2.2	1.5	1.0	1.1	0.9		
	PrecMax	27.6	3.5	2.0	2.7	4.0	6.2	8.7	7.2	6.1	7.4	4.3	2.3	2.6		
	PrecMin	12.6	0.1	0.0	0.0	0.4	0.5	0.7	0.9	0.4	0.0	0.1	0.1	0.0		
	PrecStd	4.3	0.6	0.6	0.6	1.0	1.5	1.6	1.3	1.5	1.6	1.0	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	59.5	68.0	76.5	84.1	88.6	91.3	92.3	87.5	76.3	66.2	52.8		
		MCWB	41.2	49.3	52.3	59.6	64.3	71.5	70.7	69.9	66.2	61.8	55.6	47.4		
	2%	DB	42.2	52.3	61.0	72.6	80.5	85.2	87.4	89.0	83.1	70.7	60.0	48.4		
		MCWB	39.3	46.1	49.5	57.2	63.9	69.2	70.0	68.2	63.9	59.4	52.8	44.2		
	5%	DB	39.5	47.1	56.6	68.7	77.2	82.7	84.4	86.1	79.5	66.0	55.2	43.7		
		MCWB	37.3	42.7	46.8	55.0	62.3	67.9	68.6	67.5	62.5	57.3	50.8	40.8		
	10%	DB	36.7	43.4	52.6	64.5	73.8	80.2	81.5	82.6	75.1	61.9	51.3	40.0		
		MCWB	35.1	40.5	44.8	53.4	60.4	67.0	67.9	67.0	61.5	55.4	48.2	37.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	50.2	53.4	61.0	68.8	74.5	74.8	74.4	68.3	64.5	57.0	49.1		
		MCDB	44.0	58.9	65.2	74.9	79.6	84.7	83.6	86.7	84.4	73.7	63.6	52.4		
	2%	WB	40.2	46.1	50.3	58.4	65.7	71.8	72.2	71.7	66.0	61.2	54.0	44.6		
		MCDB	41.7	51.1	59.6	70.0	75.8	82.0	82.1	83.1	77.4	67.7	59.4	48.1		
	5%	WB	37.5	43.6	47.9	56.1	63.8	70.0	70.7	69.6	64.3	58.6	51.5	41.4		
		MCDB	39.1	46.9	55.2	66.0	74.4	79.6	80.5	80.9	75.2	64.5	54.7	43.7		
	10%	WB	35.1	40.9	45.5	54.0	61.9	68.3	69.3	68.0	62.6	55.8	48.7	38.2		
		MCDB	36.5	43.2	50.9	62.9	71.3	77.4	78.6	79.5	73.1	60.9	51.2	39.8		
(35) Mean Daily Temperature Range	MDBR		10.3	12.7	16.1	18.6	19.7	19.7	20.1	21.5	19.8	17.1	11.4	10.4		
	5% DB	MCDBR	13.6	18.7	24.2	26.4	25.8	24.1	25.3	27.9	26.6	24.7	19.5	15.2		
		MCWBR	10.3	12.3	13.8	13.0	11.3	10.0	10.1	10.1	10.0	13.8	11.9	11.5		
	5% WB	MCDBR	12.2	17.9	22.5	23.8	23.3	22.2	21.8	23.6	23.6	22.1	17.3	14.6		
		MCWBR	9.7	12.5	13.4	12.2	11.1	10.6	10.2	10.0	9.8	13.0	11.2	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.350	0.388	0.448	0.421	0.418	0.444	0.452	0.408	0.373	0.345	0.318		
	taud		2.297	2.245	2.211	2.076	2.230	2.273	2.208	2.177	2.260	2.340	2.358	2.334		
	Ebn at Noon		238	254	261	255	268	269	260	252	252	244	227	224		
	Edh at Noon		24	32	38	48	43	41	43	43	36	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		391	633	1013	1397	1786	1874	1886	1709	1207	773	403	307		
	RadStd		53	84	104	124	151	165	149	128	132	93	36	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (46 neighbors)	+1.21	N/A	N/A	+1.42	N/A	N/A	-252	-360	+182	+68

Nomenclature: See separate page